Internship Report

On

Full-Stack Web Development and AI/ML Model Research & Training at ILPedia.com

Submitted To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement for the award of the degree of

Bachelor of Technology in Computer Science and Engineering

By

Name of Student: UMAR IQBAL

Roll Number: 2410030933

Program: B.Tech. CSE (Spec. in Data Science and Big Data Analytics)

Semester: 5th | Section: 3CSE12

Batch: 2026-27

CANDIDATE'S DECLARATION

I, Umar Iqbal, do hereby declare that the internship report titled “Full-Stack Web Development and AI/ML Model Research & Training at ILPedia.com” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Signature: ______________________

Student Name: Umar Iqbal

Roll No.: 2410030933

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism, and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to express my sincere gratitude to Dr. Mohammad Tariq and the entire team at ILPedia.com for granting me the opportunity to undertake this internship and for their continuous guidance, mentorship, and support throughout the two-month tenure. Their valuable insights into full-stack development and AI/ML research greatly enriched my learning experience.

I also extend my heartfelt thanks to my faculty coordinator and the School of Computer Science and Engineering, IILM University, for their constant encouragement and academic support in the successful completion of this internship and report.

I express my gratitude to my parents for their blessings.

Signature: ______________________
Student Name: Umar Iqbal
Roll No.: 2410030933

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INTERNSHIP COMPLETION CERTIFICATE

ILPedia.com

Your Gateway to the World of Ionic Liquids

Date: 28th July 2026

CERTIFICATE OF INTERNSHIP COMPLETION

This is to certify that
Mr. Umar Iqbal
has successfully completed his **two-month internship** with **ILPedia.com** from **28 May 2026 to 28 July 2026**.

During the internship, he worked in the capacity of:
Full Stack Developer with AI/ML Model Researcher & ML Model Trainer

Throughout the internship, **Mr. Umar Iqbal** contributed to practical and research-oriented projects involving **Full-Stack Web Development, Artificial Intelligence, Machine Learning, AI/ML Model Research, Model Training, Testing, and Optimization.**

His responsibilities included **developing and maintaining web applications, working on frontend and backend components, conducting research on AI/ML techniques, preparing and preprocessing datasets, training and evaluating machine learning models, performing model experimentation, integrating AI/ML solutions into applications, debugging software and models, and maintaining technical documentation.**

During his tenure, **he demonstrated technical proficiency, problem-solving ability, research aptitude, commitment, teamwork,** and a willingness to learn and explore emerging technologies.

We appreciate his contributions to the organization and wish him continued success in his academic, professional, and future endeavors.

Issued on: **28 July 2026**

Mohammad Tariq

Dr. Mohammad Tariq

contact@ilpedia.com

ILPedia.com
Your Gateway to the World of Ionic Liquids
Date: 28th July 2026

LETTER OF INTERNSHIP COMPLETION & EXPERIENCE

To Whom It May Concern,

This is to formally certify that **Mr. Umar Iqbal** has successfully completed a **two-month internship** with **ILPedia.com** from **28 May 2026 to 28 July 2026.**

During his internship, he served as a:
Full Stack Developer with AI/ML Model Researcher & ML Model Trainer

During his tenure with **ILPedia.com**, **Mr. Umar Iqbal was involved in technical and research-oriented activities spanning Full-Stack Development, Artificial Intelligence, Machine Learning, AI/ML Model Research, and Machine Learning Model Training.**

His key responsibilities and areas of contribution included:

1. Developing, maintaining, and enhancing full-stack web applications and software components.
2. Designing, developing, testing, and integrating frontend and backend features.
3. Conducting research on Artificial Intelligence and Machine Learning models, techniques, frameworks, and applications.
4. Assisting with the collection, preparation, cleaning, and preprocessing of datasets for machine learning applications.
5. Training, testing, evaluating, and optimizing Machine Learning models.
6. Conducting model experimentation and comparing different algorithms and approaches.
7. Implementing and integrating AI/ML models into practical software applications where required.
8. Conducting technical research and preparing documentation for assigned projects.
9. Debugging, troubleshooting, and improving existing applications and machine learning models.
10. Collaborating with team members to understand requirements and execute assigned technical tasks within defined timelines.
11. Maintaining documentation related to development work, research activities, experiments, and project progress.
12. Exploring emerging technologies and contributing suggestions for improving ongoing projects and technical solutions.

Mohammad Tariq

Dr. Mohammad Tariq

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CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

This report presents an account of the two-month Summer Internship undertaken at ILPedia.com from 28 May 2026 to 28 July 2026, as part of the curriculum requirement for the degree of Bachelor of Technology in Computer Science and Engineering (Specialization in Data Science and Big Data Analytics) at IILM University, Greater Noida. During the internship, I was appointed as a Full Stack Developer with AI/ML Model Researcher & ML Model Trainer, a role that combined practical, hands-on software engineering with research-oriented work in Artificial Intelligence and Machine Learning.

The internship provided an opportunity to work at the intersection of full-stack web development and applied machine learning, allowing me to strengthen my technical foundation while contributing to real, project-based assignments. This report documents the organization, the nature of the work undertaken, the objectives pursued, the technologies used, the overall system approach, the methodology followed, and the outcomes achieved during the tenure of the internship.

4.2 Organization Profile

ILPedia.com is a knowledge and technology-driven organization that describes itself as "Your Gateway to the World of Ionic Liquids," reflecting its core focus on building and disseminating research-oriented digital resources in the domain of ionic liquids and related chemical sciences. Alongside this core knowledge platform, the organization undertakes technology-driven initiatives spanning full-stack web development, Artificial Intelligence, and Machine Learning, applying these modern computational tools to research, data analysis, and digital platform development.

The organization is led by Dr. Mohammad Tariq, under whose guidance and mentorship interns are given practical exposure to both software development and AI/ML research workflows. The work culture at ILPedia.com emphasizes a blend of academic rigor and applied engineering, encouraging interns to contribute meaningfully to ongoing projects while developing professional skills in research, documentation, and collaborative development.

4.3 Problem Statement

Organizations that operate research-oriented digital platforms often face the dual challenge of maintaining robust, user-friendly web applications while simultaneously exploring the application of Artificial Intelligence and Machine Learning to enhance data processing, analysis, and decision-making. There is a need for skilled contributors who can develop and maintain full-stack web solutions and, in parallel, conduct research on ML models, prepare and process datasets, and train and evaluate models that can be integrated into practical applications. This internship was structured to address that need by involving the intern in both software development and AI/ML research and training tasks within the organization's ongoing projects.

4.4 Project Objectives

- To gain practical, hands-on experience in full-stack web development, including frontend and backend design, development, testing, and integration.
- To conduct research on Artificial Intelligence and Machine Learning models, techniques, and their real-world applications.
- To assist in the collection, cleaning, and preprocessing of datasets used for machine learning tasks.
- To train, evaluate, test, and optimize Machine Learning models, comparing different algorithms and approaches.
- To implement and integrate AI/ML models into practical, usable software applications.
- To strengthen debugging, troubleshooting, and problem-solving skills through real project work.
- To develop collaborative, teamwork, and technical documentation skills while working within organizational timelines and guidelines.

4.5 Scope of the Project

The scope of the internship work spanned the full lifecycle of software and machine learning development activities assigned by the organization. On the development side, this included designing, building, and maintaining full-stack web applications and related software components, covering both frontend user interfaces and backend services. On the research and machine learning side, the scope covered dataset preparation, model experimentation, training, evaluation, and optimization, along with the integration of trained models into functional applications.

The work was project-based and research-oriented, meaning that tasks evolved according to the organization's ongoing requirements. The scope also included maintaining proper documentation of development work, research findings, and experiments, as well as exploring emerging technologies and suggesting improvements to existing projects — ensuring that the internship contributed both immediate technical value to the organization and a broad learning experience to the intern.

4.6 Technologies and Tools Used

The internship involved exposure to and use of a range of technologies and tools commonly used in full-stack development and applied machine learning, including:

- Frontend Technologies: HTML5, CSS3, JavaScript, and modern frontend frameworks/libraries for building responsive user interfaces.
- Backend Technologies: Server-side frameworks and RESTful API design for building and integrating application logic.
- Databases: Relational and/or NoSQL databases for structured and unstructured data storage.
- Machine Learning & AI: Python-based ML libraries and frameworks used for data preprocessing, model training, evaluation, and experimentation.
- Version Control & Collaboration: Git/GitHub for source control, along with collaborative documentation practices.
- Development Tools: Code editors, debugging tools, and testing utilities used throughout the development and research cycle.

4.7 System Architecture

The overall system approach followed during the internship was based on a standard layered architecture typical of full-stack web applications integrated with machine learning components. At a high level, the architecture can be described as follows:

- Presentation Layer: The frontend interface through which users interact with the application.

- Application/Backend Layer: Server-side logic that handles requests, business rules, and communication between the frontend and the data/ML layers.
- Data Layer: Databases and datasets used for storing application data as well as training and evaluation data for machine learning models.
- ML/AI Layer: Trained machine learning models integrated into the backend, responsible for performing predictions, classifications, or other AI-driven tasks as required by the application.

This layered approach ensured a clear separation of concerns between application development and machine learning research, allowing both streams of work to be developed, tested, and integrated in a structured and maintainable manner.

4.8 Methodology

The work during the internship followed an iterative and collaborative methodology, consistent with agile development practices. Tasks were assigned based on project requirements, and progress was reviewed periodically under the guidance of the mentor. The general methodology followed for development tasks involved requirement understanding, design, implementation, testing, and integration of frontend and backend components.

For machine learning tasks, the methodology followed the standard ML pipeline: data collection and preprocessing, exploratory analysis, model selection, training, evaluation using appropriate metrics, and iterative optimization through experimentation with different algorithms and hyperparameters. Throughout both streams of work, proper documentation was maintained, and debugging and troubleshooting were carried out to continuously refine both the applications and the models.

4.9 Expected Outcomes

By the end of the internship, the following outcomes were achieved:

- Practical, working knowledge of full-stack web development, including frontend and backend integration.
- Hands-on experience in preparing datasets and training, evaluating, and optimizing machine learning models.

- Improved ability to research, compare, and apply different AI/ML techniques to practical problems.
- Strengthened debugging, testing, and software engineering skills through real project involvement.
- Enhanced professional skills, including teamwork, technical documentation, and adherence to project timelines.
- Successful completion of the internship, as formally recognized through the Certificate of Internship Completion and the Letter of Internship Completion & Experience issued by ILPedia.com.

4.10 Certificates of Completion and Other Communication/Mail Proof

The following documents, issued by ILPedia.com, are attached as proof of the successful completion of this internship:

- Letter of Appointment (dated 28 May 2026).
- Certificate of Internship Completion (issued 28 July 2026).
- Letter of Internship Completion & Experience (issued 28 July 2026).

The Internship Completion Certificate has been included earlier in this report (see Chapter 3). Copies of the Letter of Appointment and the Letter of Internship Completion & Experience are appended as supporting documentation.

CHAPTER 5: BIBLIOGRAPHY / REFERENCES

1. ILPedia.com – Letter of Appointment, issued by Dr. Mohammad Tariq, 28 May 2026.

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8. IILM University – Internship Report Format Guidelines, School of Computer Science and Engineering, 2026-27.